(i.e) a line parallel to the bar magnet). Turn the magnet gently and let it come to stop again. Repeat it three or four times.

Does the bar magnet stop in the same direction each time? _____

In which direction does the magnet stop every time? _____

This is roughly the north-south direction. The end of the magnet that points to the north is called the **North Pole.** The end that points to the south is called the **South Pole.**

A freely suspended magnet always comes to rest in north-south direction.



The directive property of magnets has been used for centuries to find directions. Around 800 years ago, the Chinese discovered that a suspended lode stone stops in the north-south direction. Chinese used these lode stones to find directions.

The navigators of that country used to keep a piece of lode stone suspended in their boats and during a storm or mist, they used the lode stone to locate directions.



1.6 Magnetic compass

A compass is an instrument which is used to find directions. It is mostly used in ships and airplanes. As a rule, mountaineers also carry a compass with them so that they do not lose their way in unknown places.



The compass has a magnetic needle that can rotate easily. The marked end of the needle is the North Pole of the magnet. **Can you use magnetic compass to find west direction?** Ask your teacher to help you in using magnetic compass.

1.7 Properties of Magnets

Attraction or Repulsion

Take two similar magnets, place them in four different ways as shown in Figure.

